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Dynamic performance of CAES based on switching expansion reduction

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Abstract

Compressed air energy storage (CAES) is considered as one of the most promising large scale energy storage technologies for the electrical grid with high penetration of renewable energy. CAES systems are required to operate under complex conditions because of the pressure change in the air storage chamber and input/output power changes considering the fluctuated renewable generation and the mismatch between energy supply and demand. A dynamic operation configuration was proposed to satisfy the adjustment of off-design conditions of CAES systems and frequency management auxiliary service market. To regulate the inlet pressure of turbines and meet the output power demand for better performance, a thermodynamic model of a CAES discharging system with thermal storage and the pressure control unit (SER) was established. The control strategy of setting adjustment width instead of single instruction curve is proposed. Then, the dynamic operation parameters including pressure, mass flow rate, and exergy change during discharging processes were investigated. The superiority of the control accuracy and efficiency was demonstrated over single throttle valve through valve characterization studies. The SER system improves control accuracy by decoupling pressure and flow rate changes in dynamic operation, solving the problem of difficult control and low accuracy of traditional throttle valves. The expansion tank in SER serves as a pressure buffer and heat exchanger with the environment, reducing energy loss during fluid flow. Furthermore, the effect of the expansion tank volume of switch expansion reduction in exergy destruction were assessed. For a 10 MW/60MWh system with dynamic load to meet the requirements of the unit to participate in the frequency modulation auxiliary service market, the optimal added expansion tank volume is about 70m³. The optimization method of control strategy of SER system is also explored. The proposed optimized SER system of volume and control strategy can smoothly regulate the inlet pressure of the expander. When the adjustment width is 3bar, the frequency is 50% decreased, meanwhile, the on-off action time is more evenness, which ensures the stable and efficient operation of the CAES system and improves the comprehensive performance of the system.

Keywords Compressed air energy storage system, Discharge process, Dynamic operation, Pressure control, Exergy analysis, Optimization strategy

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1 Introduction

To actively address global warming, nations worldwide have committed to lowering their carbon dioxide emissions, with some even putting forward a clear timetable for the carbon neutrality. The large-scale development and deployment of renewable energy such as wind energy and solar energy is an important means to constitute decarbonized energy systems [1–3]. Energy storage is a key technology to solve the large-scale consumption and adjust output of renewable energy and improve the efficiency and security of the power system [4–7]. In bulk power system networks, the Independent System Operator (ISO) has the responsibility to maintain the target grid frequency and scheduled tie flows. The dual objectives mentioned above are achieved by means of Automatic Generation Control (AGC) aiming to minimize the Area Control Error (ACE) by re-dispatching the committed generation units via the ancillary service market. With ever-increasing penetration of intermittent generation resources in the electric grids, the ISOs are facing challenges to adjust the power accordingly and meet the system AGC requirements. In such an environment, the "non-generation" resources like Energy Storage Systems (ESS) are participating in regulatory markets on a non-discriminatory basis, assumes significant importance. The ESS has been often referred to as one of the potential solutions to substantially improve the power adjustment and frequency response especially in systems characterized with high renewable energy penetration levels [8].

Electric energy storage (EES) technologies include pumped-hydro storage, compressed air energy storage, flywheel storage, lithium battery, flow battery, superconducting magnetic energy storage, supercapacitor energy storage, etc. [9]. Among them, the compressed air energy storage (CAES), with high efficiency, large capacity, long storage time and low unit cost, is considered to be one of the most prospective large-scale energy storage technologies. In order to effectively improve the performance of CAES systems, researchers have proposed many types of CAES systems such as adiabatic compressed air energy storage (ACAES), compressed air energy storage with thermal storage (TS-CAES), isothermal compressed air energy storage (I-CAES), liquid air energy storage (LAES), supercritical compressed air energy storage, underwater compressed air energy storage and some others [10]. TS-CAES has the advantages of no fossil fuel consumption, simple process and high efficiency, and has attracted wide attention [11].

Generally, the storage vessel or cavern is isochoric. Thus, the air conditions in the storage cavern keep changing during charging and discharging processes [12]. Secondly, when the CAES is connected to the grid, the power supply and demand are also changing [13,

14]. Furthermore, the CAES system can also participate in the frequency management (FM) market service which requires the energy storage system to have better dynamic performance.

Thus, the CAES system discharging process is always under dynamic operations [15]. It is necessary to adjust the high-pressure air flowing out of the air storage cavern effectively to match the inlet pressure of the expander [16].

At present, the throttle valve is usually utilized to adjust the released air from the storage to the inlet of the expander, which makes the expander run close to the design condition and can better ensure the rated output power [17, 18]. However, the throttle valve is essentially a throttling element with variable local resistance. By changing the throttling area, the gas flow rate and fluid kinetic energy are adjusted. Due to certain pressure loss caused by the friction of the gas in the flow process, the pressure of the throttle valve is reduced. The flow equations of a valve indicated the relationship of the flow rate, inlet pressure and outlet pressure. This makes it difficult to accurately control the output pressure after the valve through the throttle valve [12, 19].

Researchers have explored the pressure control modes between the air storage cavern and the expander including the throttle valve, the combination of throttle valve and injector, the guided vane adjustment [12]. Zhao et al. [20] compared energy loss between sliding pressure mode and constant pressure mode using throttle valve during CAES discharging process, and found that the roundtrip efficiency of sliding pressure mode is a little higher (45.16%) than the constant inlet pressure mode (44.56%). Guo et al. [21] compared the effects of throttling decompression and injector decompression respectively on the overall system performance during the energy release process of the CAES system, and the results showed that the energy storage efficiency after the use of injector was increased from 61.95% to 65.36%. Guan et al. [22] carried out the orthogonal experimental design on the inlet pressure of nozzle under different base pressure. Based on the response surface method, the optimal nozzle governing method (NG) corresponding to different base pressure under required output work conditions is obtained by multi-island genetic algorithm. The optimal NG method can improve the efficiency by up to 9.2% compared with throttle governing. Further, Guan et al. [23] integrates nozzle control (NC) and stator mounting angle (RSIA) adjustment, optimizes compressor specific power (+6.1%), analyzes turbulence dissipation's dominant role in losses (88.7%), and proposes pressure-adaptive strategies to expand system operation range. Qing et al. [24] analyzed parameters sensitiveness of primary fluid pressure, second fluid pressure and so

on for a 10 MW TS-CAES system under constant pressure operation and the results showed that the roundtrip efficiency is increased by 10% with the ejector adjustment compared that without ejector. It was also found that the low-pressure air source and motive air pressure of the ejector should be carefully selected owing to their significant influence on the performance of TS-CAES systems. Chen et al. [25] designed three single-stage bypass systems to regulate turbine inlet and outlet parameters, extending power generation duration (up to 6.06 h), improving energy density (maximum 16.04 MJ/m³), and optimizing performance under variable pressure conditions. Guo et al. [26] studied optimal adjustment of inlet vanes of a 4-stage compressor considering changing back pressure and mass flow rate for a 10MW ACAES system, and found that the exergy efficiency can be improved by 2.5% compared with that of compressor without guided vane adjustment. Wang et al. [27] developed a nonlinear thermodynamic model and proposed an iterative solution framework (MP-SP) to enhance optimization speed (100 times faster than conventional methods), ensuring safe operation and improving economic efficiency and reliability. Xu et al. [28] established a micro-CAES system test platform to investigate the performance of scroll expanders and generators under varying parameters. By optimizing compressor-generator matching, generator efficiency achieved up to 95%. Simone et al. [29] built a dynamic model of a small-scale CAES and studied the control strategy and the capability of the model of the proposed plant to follow a variable power profile, and found tense oscillations due to the integral component of the throttling valve's PI controller that directly effects the air mass flow rate, causing accuracy worsens. Yang et al. [30] indicated that throttle valves, compressors, and turbine expanders are the main sources of exergy loss in medium-temperature TS-CAES systems (VLV2 accounting for 18.7%, compressors 35.5%, turbine expanders 30.3% of total exergy loss), emphasizing that optimizing throttle valve design to reduce pressure loss during constant-pressure discharge is a key strategy. They also proposed optimizing compression/expansion processes and improving heat exchanger efficiency to further cut exergy loss and boost system performance. Calise et al. [31] coupled MATLAB and TRNSYS to model a hybrid renewable-CAES system, using real compressor and turbine maps to keep compressors at ~81% and turbines at ~80% isentropic efficiency and reach 59% annual round-trip efficiency. They recovered 54% of compression heat with an oil loop and cooled the compressor inlet with a water loop, while 2 000 m² of solar collectors supplied 47% of turbine preheat. A heat exchanger kept the tank at 20 °C during discharge, and smart compressor control minimized grid exports, letting the system cover 14% of

district demand and provide 23% of local self-consumption. Yun et al. [32] proposed an SOFC-CAES integrated system (CAES with "dual compressors+dual coolers") linked to SOFC's power and waste heat (15%–20% higher efficiency), and used regression machine learning ($R^2 > 98\%$) to optimize CAES, reducing compressor energy use by 8%–10%, limiting tank pressure fluctuation to $\pm 3\%$, and raising CAES net efficiency to 51%, providing references for CAES component optimization and dynamic regulation.

Although previous studies have examined variable operating conditions under pressure changes, the impact of valves on dynamic characteristics has received little attention.

In applications such as renewable energy generation with high variability and grid regulation, the CAES system requires precise and rapid control of pressure differential and flow between the storage chamber and expander, particularly under continuous dynamic changes during discharge process. The flow characteristics of a throttle valve indicated that it was difficult to regulate the throttle valve in a precise way to fulfil the flow rate demand for a large adjustment flow rate as the mass flowrate is coupled with the inlet/outlet pressure of the throttle valve [12]. Switching expansion reduction (SER) uses an electrical switching valve to realize electronically controlled pressure reduction for high pressure pneumatics [12, 33, 34]. SER is relatively easier to realize electrically controlled pressure reduction, and precise to regulate the air pressure in a large range. Furthermore, on–off controller is utilized for SER instead of analog controller, the control strategy is relatively simple. With respect to the pressure and power control requirement, further to the above-mentioned method, a dynamic regulation and control strategy of the discharge process by combining valves and expansion tank is proposed [12, 35, 36]. A simulation study was carried out to simulate the performance of SER in CAES system. The control strategy was performed and the flow rated was controlled precisely in a large range. This device also decreased the pressure loss compared with throttling valve. The exergy destruction was decreased by 15 kJ/kg, and the roundtrip efficiency was increased by 0.2% compared with that of CAES with throttle valve adjustment [12, 35, 36].

The first integration of a switching expansion reduction (SER) unit into a full-system dynamic model of a thermal storage compressed air energy storage (TS-CAES) system is proposed, focusing on AGC-style load-following performance. Combined with the application requirements of stabilizing renewable energy generation and participating in the AGC (Automatic Generation Control) frequency modulation service market, the working performance of pressure, mass

flow and exergy destruction, and control methods of the system under dynamic operation conditions are explored. A systematic comparison of SER with single-throttle valve control under both rated and dynamic (AGC) conditions. The first exergy-optimal sizing of the SER expansion tank for dynamic CAES operation. At the same time, a new expansion tank parameter control strategy is proposed to meet the requirements of dynamic change accuracy of the system. This paper is aimed to provide a theoretical reference for adjusting the CAES power output precisely and promptly, and reducing irreversible loss and regulating the air pressure as required.

This article established a CAES discharging system using Switching Expansion Reduction unit to regulate the inlet mass flow and pressure of turbine. Section 2 describes the system configuration; Section 3 introduces the simulation methodology of essential parts in discharging process of the CAES system and SER system. Section 4 introduces the major evaluation indexes representing the system characteristic and the effects of parameters such as expansion tank volume on the energetic or dynamic performance. Section 5 explores the optimal strategy of SER system to reduce energy loss and improve dynamic performance. Finally, Section 6 summarizes the main conclusions.

2 System description

2.1 The CAES system

The schematic diagram of the TS-CAES system based on Switching Expansion Reduction (SER) is illustrated in Fig. 1. The proposed system consists of a four-stage compression, four-stage expansion, heat exchangers, a compressed air storage cavern and a SER unit instead of single throttle valve. In charging process, the air is compressed by compressors and the heat of compression is absorbed and stored. In discharging process, the high-pressure air passes through the SER control unit and its pressure are regulated to the required pressure, then enters into the expander to generate electricity after being reheated.

2.2 Switching Expansion Reduction system

SER as shown in Fig. 2 consists of a switch valve, a regulation valve and an expansion tank. The air mass flowing into the expansion tank is controlled by the on-off time of the switch valve, and the air pressure is reduced by compressed air expanding in the expansion tank. The air in the expansion tank is then regulated to a certain pressure to enter the expander for power generation [33, 37, 38]. Compared with the single-throttle-valve mode, this proposed system of SER decouples the pressure difference of valve inlet and outlet and the mass flow rate. Compared to systematic adjustment methods such as

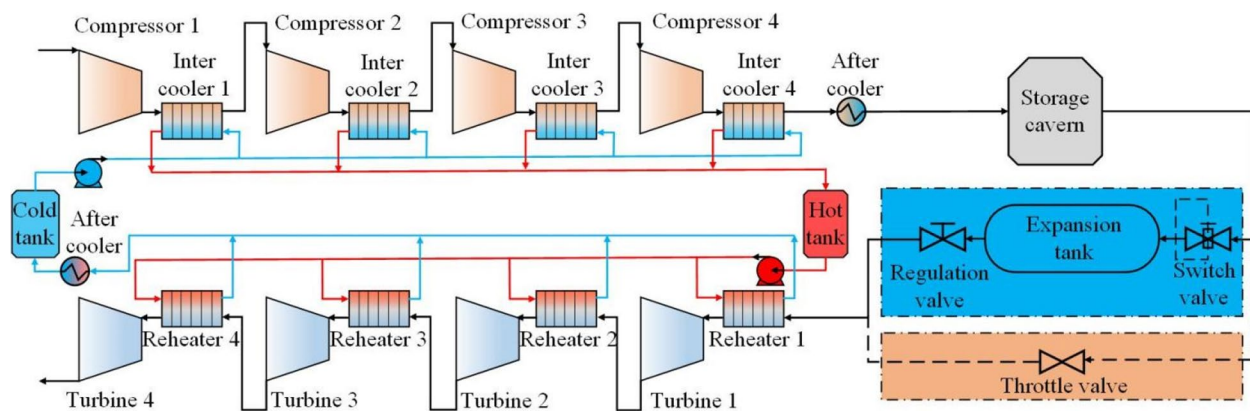


Fig. 1 System configuration of a TS-CAES system based on SER

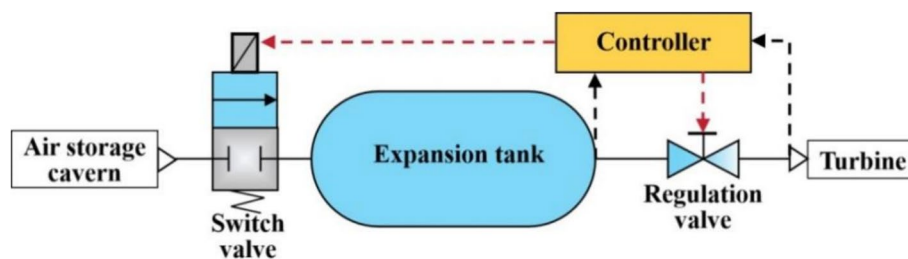


Fig. 2 Schematic diagram of switching expansion reduction (SER) pressure control unit

variable guide vanes, variable speeds, etc., SER offers higher response speeds and simpler regulation [12, 34].

It is assumed that heat exchanges occur between the inside and outside of the expansion tank. In simulation, heat transfer coefficients are calculated by these equations in reference [39]. The signals of the pressure sensors in the expansion tank and the inlet of turbine are transmitted to the controller, which control the on-off time of the switch valve and the regulation valve opening area. The air storage cavern experiences an intermittent discharging process to the expansion tank, while the expansion tank undergoes an intermittent charging and continuous discharging processes. The regulation valve is used to further adjust the air flowrate to satisfy the demand.

3 Methodology

3.1 CAES System

3.1.1 Expanders

When the expander operates under off-design conditions, its expansion ratio and efficiency vary with different rotation speeds, flow rates and other parameters. Off-design performance modelling of the expander is considered where a modified Flügel formula is used [40]. The modified Flügel formula integrates theoretical analysis with experimental patterns to build the model and applies mathematical methods to determine the correction coefficients from the data [41].

$$\dot{G}_e = \sqrt{1.4 - 0.4\dot{n}_e} \sqrt{(1/\pi_e^2 - 1)/(1/\pi_{e0}^2 - 1)} \quad (1)$$

$$\dot{n}_e = [1 - t_4(1 - \dot{n}_e)^2] (\dot{n}_e/\dot{G}_e) (2 - \dot{n}_e/\dot{G}_e) \quad (2)$$

$$t_4 = 0.3 \quad (3)$$

$$\dot{G}_e = (G_e \sqrt{T_{e,in}/p_{e,in}}) / (G_e \sqrt{T_{e,in}/p_{e,in}})_0 \quad (4)$$

$$\dot{n}_e = (n_e / \sqrt{T_{e,in}}) / (n_e / \sqrt{T_{e,in}})_0 \quad (5)$$

$$\dot{\eta}_e = \eta_e / \eta_{e0} \quad (6)$$

where, $\dot{G}_e$ is relative reduced mass flow rate; $\dot{n}_e$ is relative reduced rotation speed; π_e, π_{e0} are expansion ratio and design expansion ratio; G_e is air mass flow rate; n_e is rotation speed; η_e is isentropic efficiency; $p_{e,in}, T_{e,in}$ are inlet pressure and inlet temperature of turbine; subscript 0 represents designed value.

The outlet temperature, outlet pressure and power output of each stage are calculated as shown below (7)-(9):

$$T_{e,out} = T_{e,in} \left[1 - \left(1 - \pi_e^{\frac{k-1}{k}} \right) \eta_e \right] \quad (7)$$

$$p_{e,out} = p_{e,in} / \pi_e \quad (8)$$

$$P_e = G_e (h_{e,in} - h_{e,out}) = G_e c_p (T_{e,in} - T_{e,out}) \quad (9)$$

where, $p_{e,out}, T_{e,out}$ are outlet pressure and outlet temperature of turbine; $h_{e,in}, h_{e,out}$ are inlet and outlet specific enthalpy of turbine; c_p is the specific heat at constant pressure; P_e is the expansion power of turbine.

3.1.2 Air storage cavern/expansion tank

The air mass flow and energy flow between the air storage cavern and the expansion unit undergo an unsteady thermodynamic process. The expansion tank is the same as the air storage cavern.

Mass and energy conservation equations are shown below (10):

$$\begin{cases} \frac{dm_{av}}{dt} = \dot{m}_{av,in} - \dot{m}_{av,out} \\ \frac{d(m_{av} u_{av})}{dt} = \dot{m}_{av,in} h_{av,in} - \dot{m}_{av,out} h_{av,out} - UA(T_{av} - T_{env}) \end{cases} \quad (10)$$

where, $\dot{m}_{av,in}, \dot{m}_{av,out}$ is the mass flow rate of charging and discharging process of the air storage cavern or the expansion tank; m_{av} is the mass in the vessel; t is time; u_{av} is specific internal energy; T_{av} is air temperature in the vessel; T_{env} is environment temperature; U is heat transfer coefficient between the vessel and environment; A is the heat transfer area.

The following equations are combined with energy conservation equations to calculate the air state in the cavern and expansion tank.

$$\begin{cases} \frac{dp}{p} + \frac{dV}{V} = \frac{dT}{T} + \frac{dm}{m} \\ \frac{dT_{av}}{dt} = \frac{1}{m_{av}} \left(\dot{m}_{av,in} \frac{c_p}{c_v} T_{av,in} + \left[\left(1 - \frac{c_p}{c_v} \right) \dot{m}_{av,out} - \dot{m}_{av,in} \right] T_{av} \right) - \frac{UA}{m_{av} c_v} (T_{av} - T_{env}) \\ \frac{dp_{av}}{dt} = \frac{R_g}{V_{av}} \frac{c_p}{c_v} (\dot{m}_{av,in} T_{in} - \dot{m}_{av,out} T_{av}) - \frac{UA R_g}{V_{av} c_v} (T_{av} - T_{env}) \end{cases} \quad (11)$$

where, c_v is the specific heat at constant volume; R_g is ideal gas constant; V_{av} is the volume of cavern or expansion tank.

3.1.3 Heat exchangers

The heat exchanger is used for interstage cooling (inter-cooler) of multistage compressor and interstage reheat (reheater) of multistage expander in TS-CAES system. Active heat storage is adopted, and water is selected as the circulating medium [42].

According to the conservation of energy, the heat released by air is equal to the heat absorbed by circulating water.

$$\dot{m}_{hot}(h_{hot,in} - h_{hot,out}) = \dot{m}_{cold}(h_{cold,out} - h_{cold,in}) \quad (12)$$

where, the *cold* and *hot* represent cold fluid and hot fluid; *in* and *out* is inlet and outlet of heat exchanger.

Counter flow heat exchangers are modelled by employing ε -NTU method [26]:

$$\begin{aligned} \varepsilon &= \frac{\dot{m}_{hot}c_{p,hot}|T_{hot,in} - T_{hot,out}|}{(\dot{m}c_p)_{min}(T_{hot,in} - T_{cold,in})} = \frac{\dot{m}_{cold}c_{p,cold}|T_{cold,out} - T_{cold,in}|}{(\dot{m}c_p)_{min}(T_{hot,in} - T_{cold,in})} \\ (\dot{m}c_p)_{min} &= \min[\dot{m}_{hot}c_{p,hot}, \dot{m}_{cold}c_{p,cold}] \\ \chi &= \frac{(\dot{m}c_p)_{min}}{(\dot{m}c_p)_{max}} \\ \varepsilon &= \frac{1 - \exp[-NTU(1 - \chi)]}{1 - \chi \exp[-NTU(1 - \chi)]} \\ NTU &= \frac{U_h A_h}{(\dot{m}c_p)_{min}} \end{aligned} \quad (13)$$

where, ε is effectiveness of heat exchanger; NTU is number of heat transfer units; χ is the ratio of heat capacities of two fluids; U_h is heat exchanger coefficient; A_h is heat exchanger area.

The outlet temperature of air $T_{air,out}^{i,e}$ and water $T_{water,out}^{i,e}$ in each exchanger during discharging process can be calculated by following Eq. (14):

$$\begin{cases} T_{air,out}^{i,e} = T_{air,in}^{i,e} + \varepsilon \frac{(\dot{m}c_p)_{min}}{(\dot{m}c_p)_{air}} (T_{water,in}^{i,e} - T_{air,in}^{i,e}) \\ T_{water,out}^{i,e} = T_{water,in}^{i,e} + \varepsilon \frac{(\dot{m}c_p)_{min}}{(\dot{m}c_p)_{water}} (T_{water,in}^{i,e} - T_{air,in}^{i,e}) \end{cases} \quad (14)$$

The water came out from each stage and mixed together

$$\sum_{i=1}^N m_i h_{i,in} = h_{tan k} \sum_{i=1}^N m_i \quad (15)$$

where, m_i is mass flow of each circulating water; $h_{i,in}$ is specific enthalpy of each outlet circulating water; $h_{tan k}$ is specific enthalpy of hot water in hot tank.

3.1.4 Switch valve and regulation valve

The working principle of the switch valve and regulation valve is similar to the orifice plate, and the switch valve is equivalent to the orifice plate for mathematical modelling. The mass flow rates is derived by using Bernoulli equation and continuity equation under adiabatic condition [33]:

$$\begin{cases} G_{valve} = C_q S p_1 \sqrt{\frac{k}{R_g T_1}} \phi(p_2, p_1) \\ \phi(p_2, p_1) = \begin{cases} \sqrt{\frac{2}{k-1} \cdot \left(v^{\frac{2}{k}} - v^{\frac{k+1}{k}} \right)}, v_{cr} < v \leq 1 \\ \sqrt{\frac{2}{k-1} \cdot \left(v_{cr}^{\frac{2}{k}} - v_{cr}^{\frac{k+1}{k}} \right)}, v \leq v_{cr} \end{cases} \\ v = \frac{p_2}{p_1} \end{cases} \quad (16)$$

where, G_{valve} is mass flow rate of switch valve or regulation valve; C_q is flow coefficient; R_g is ideal gas constant; k is air specific heat capacity ratio; ϕ represents a function containing pressure ratio; S is Valve effective cross section area; p_1 is upstream pressure; p_2 is downstream pressure; T_1 is upstream temperature; v is the valve operating pressure ratio; v_{cr} is critical pressure ratio, where flow rate reaches local sound velocity (c) when the outlet-to-inlet pressure ratio attains a specific value during flow from high to low pressure regions. The derivation of critical pressure ratio (v_{cr}) is based on the state equation of ideal gas and the law of conservation of energy.

$$\begin{cases} \frac{v^2}{2} + h = h^* \\ h = c_p T \\ v^2 = \frac{2k}{k-1} R_g T^* \left(1 - \left(\frac{p_2}{p^*} \right)^{\frac{k-1}{k}} \right) \\ c = \sqrt{k R_g T} \end{cases} \quad (17)$$

where, the superscript $*$ represents stagnation parameters and c is local sound velocity. Bringing the critical parameters (c_{cr} , p_{cr} , T_{cr}) into Eq. (17) can derive the critical pressure ratio.

$$v_{cr} = \left(\frac{2}{k+1} \right)^{\frac{k}{k-1}} \quad (18)$$

In the throttling state, the enthalpy before and after the throttle valve can be considered to be equal:

$$h_{valve, in} = h_{valve, out} \quad (19)$$

3.2 Evaluation indicators

The purpose of the study is to evaluate the beneficial value of the SER pressure control unit compared with the conventional single throttle valve unit and to optimize the SER unit. The indicators include the regulation accuracy and the exergy analysis illustrated below.

The inlet mass flow rate of turbine is linear to the inlet pressure [43], under which application scenario of the regulation valve or throttle valve, the flow rate characteristic is calculated based on the formula of the flow of orifice plate [33] and shown in Fig. 3. The curves illustrating the variation of mass flow rate with valve opening approximate a hyperbolic profile. When the upstream pressure is relatively high (e.g., under the 10 MPa operating condition mentioned in the manuscript), the slope of these curves becomes extremely steep. Specifically, a mere 1% change in valve opening can lead to a fluctuation in mass flow rate ranging from 5 to 8% of the control range. This high sensitivity implies that even minor deviations in regulation (such as a 0.5% error in opening control) may result in the actual flow rate deviating from the target value by more than 5%, ultimately severely compromising the accuracy of flow regulation.

A key phenomenon is that when the valve opening falls within the range of approximately 0.1 to 0.3, the curve cluster (corresponding to different pressure conditions) exhibits an obvious tendency of dense aggregation. Taking typical operating conditions within this range as an

example, if the upstream pressure fluctuates between 7 and 9 MPa, the difference in flow rate at the same opening can reach 15% to 20% of the rated flow rate. Even if the opening is precisely controlled within the range of $\pm 0.5\%$ of the target value, the actual flow rate may still exceed the allowable error range (usually $\pm 3\%$), which poses significant challenges to achieving accurate and stable regulation.

In contrast, when the pressure difference between the upstream and downstream of the valve is small, the slope of the curves flattens significantly, as shown in Fig. 3. Under such circumstances, a 1% change in valve opening can restrict the fluctuation of mass flow rate to within 2% to 3% of the control range. Moreover, within the valve opening range of 0.1 to 0.8, the dispersion of the curves under different pressure conditions is greatly reduced—the difference in flow rate at the the same opening is only 3% to 5% of the rated flow rate. This gentle slope weakens the strong coupling relationship between valve opening and mass flow rate, thereby enhancing the system's precision performance. This is of great significance for ensuring that the compressed air energy storage (CAES) discharging system meets the accuracy requirements of frequency modulation auxiliary services.

The exergy changes and destruction are also analyzed to reveal the irreversible loss of SER system. The control volume and exergy flow of SER system is shown in Fig. 4. For the total air in the initial state of SER, including the

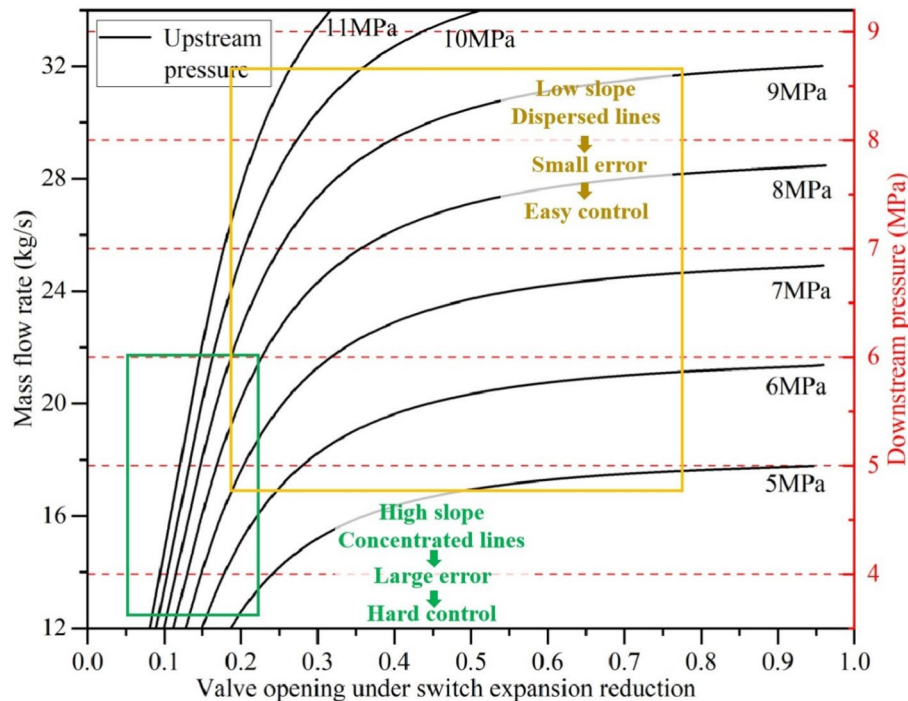


Fig. 3 Flow rate characteristics of regulation/throttle valve under expander-inlet demand

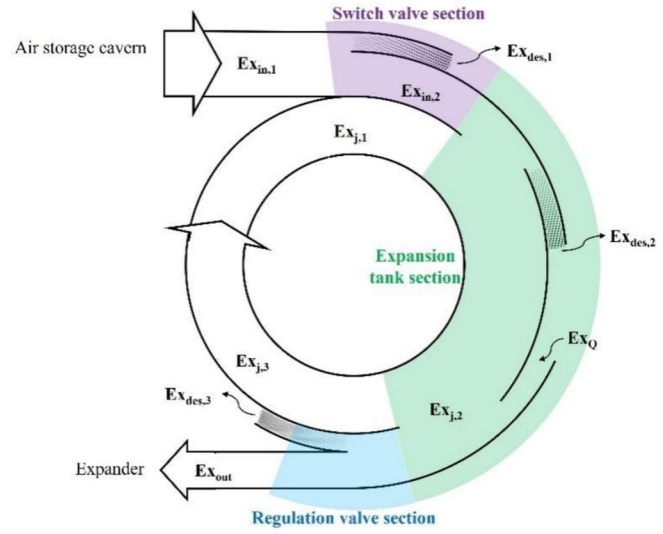
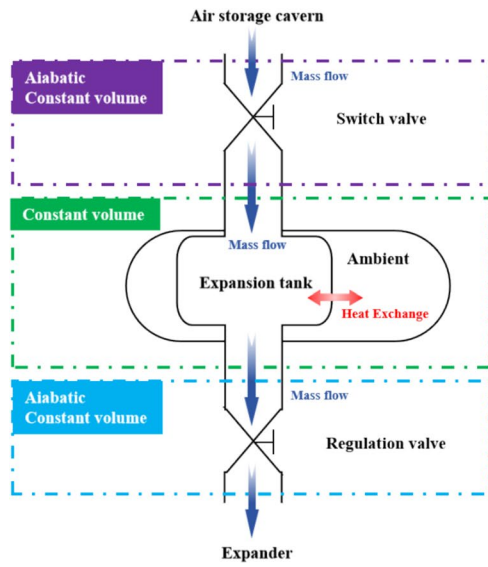


Fig. 4 **a** The control volume and **(b)** the exergy flow of SER system, in which Ex_{des} means exergy destruction; Ex_{in} and Ex_{out} is inlet and outlet exergy of SER system; Ex_j is exergy in expansion tank

air in storage cavern and expansion tank and output air, it is an expansion and pressure decreasing process for constant mass of air. The pressure interface in SER is the switch valve, in which the air flows from high pressure zone (air storage cavern) into low pressure zone (expansion tank), it is assumed to be ideal gas and the exergy change is determined by the state parameters in air storage cavern and expansion tank. The switching valve operates through adiabatic throttling, and high-pressure air passing through the valve generates irreversible exergy destruction due to vortexes and friction. For the air in expansion tank, when SER operates dynamically, the exergy change is caused by the thermodynamic process of air supplies and heat exchanges with environment. When the air passes through the throttle valve after expansion tank, it is similar to a real gas isenthalpic process so as to the switch valve.

The exergy in vessel and exergy change during flow are determined by (20)-(21):

$$Ex = m[u - u_0 - T_0(s - s_0) - p_0(v - v_0)] \quad (20)$$

$$\Delta Ex = Ex_{out} - Ex_{in} \quad (21)$$

Defining 101kPa and 298K as the reference state, in SER system, the exergy change of switch valve, expansion tank and regulation valve are calculated by:

$$\Delta Ex_{sw} = -G_{valve} T_{env} (s_{sw,out} - s_{sw,in}) \quad (22)$$

$$\Delta Ex_j = m_j [u_j - u_0 - T_{env} (s_j - s_0) - p_0 (v_j - v_0)] \quad (23)$$

$$\Delta Ex_{reg} = -G_e T_{env} (s_{reg,out} - s_{reg,in}) \quad (24)$$

where, ΔEx_{sw} , ΔEx_j , ΔEx_{reg} are the exergy changes of the switch valve, expansion tank and regulation valve; $s_{reg,in}$ and $s_{reg,out}$ are the specific entropy of the inlet and outlet flow of the regulation valve; v is the specific volume; $T_{j,ini}$ and $T_{j,ini}$ are the initial temperature and pressure of the expansion tank; subscript sw represents switch valve; j represents expansion tank; reg represents regulation valve; env is the environment condition; ini is the initial state.

The exergy loss of the SER system is defined as the negative value of the sum of exergy change of the switch valve and the regulation valve.

$$Ex_{des,total} = -(Ex_{reg,out} - Ex_{sw,in} + \Delta Ex_j) \quad (25)$$

where, ΔEx_j is exergy change in expansion tank.

Switching frequency of switch value can affect its service life and operation stability. Furthermore, the opening and closing action need reaction time and action time, resulting in the decrease of SER response speed and adjustment accuracy.

3.3 Modelling verification

In order to verify the accuracy of the modelling especially SER system, this paper compared the MATLAB simulation results with the experiments of Luo et al. [34] based on the experimental parameters in that literature. Initial pressures and temperatures in simulations are obtained from experiments. In the experimental setup, high-pressure air from supply tanks passes through an inlet

choke and a switch valve before entering the expansion tank, which is ventilated to the atmosphere. The output flow rates are regulated by the diameters of outlet choke orifices. Pressure measurements are recorded using sensors PP1 and PP2, while temperature data is obtained via K-type thermocouples PT1 and PT2 [34] (Fig. 5).

Some parameters of the experimental system are shown in Table 1 [33], two experimental situations are used to verified the modelling (Table 2).

The verification result is shown in Fig. 6 and the simulation modelling reflected the actual situation well. In situation 1 (Fig. 6a), the target of the verification system is to output constant pressure air from a constant volume tank. During the process of the decompression, the expansion tank pressure gradually builds up to the designed outlet pressure and then basically remains 2 MPa with a maximum error of 2%. At the same time, the supply pressure declines linearly at a rate of 0.09 MPa/s, consistent with the experimental discharge profile. In situation 2, due to turbulent disturbances during the throttling expansion of high-pressure gas and fluctuations in ambient heat dissipation, the measured pressure curve

showed small fluctuations of ± 0.05 MPa, but the overall trend was highly consistent with the simulation results (Fig. 6b). Both simulation and experiment showed a rapid upward trend, eventually stabilizing around 0.5 MPa.

To quantify the agreement between simulation and experiment, Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) were calculated for above two cases. The formulas and results are shown in Table 3.

4 Results and discussion

4.1 System parameters and operating condition design

4.1.1 System parameters

To study the off-design characteristics of the system, it is necessary to firstly determine the design parameters of the system, and then conduct a study of the off-design characteristics under the determined system. The design parameters of the TS-CAES system are shown in Table 4. The minimum and maximum pressures of air storage cavern are 7 MPa and 10 MPa, respectively. The rated power and discharge time of the TS-CAES system under design

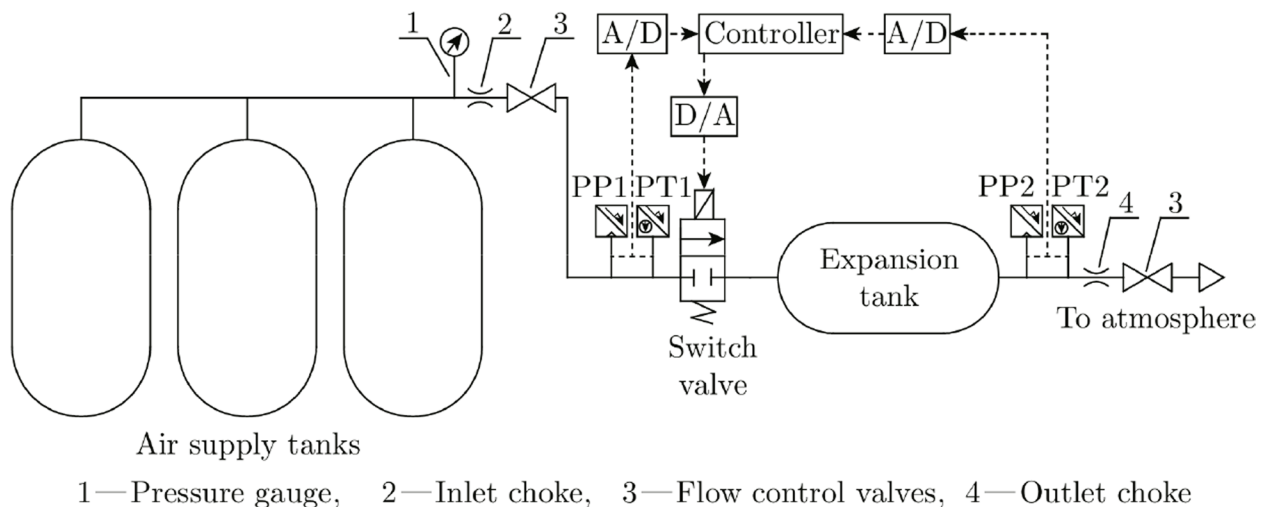


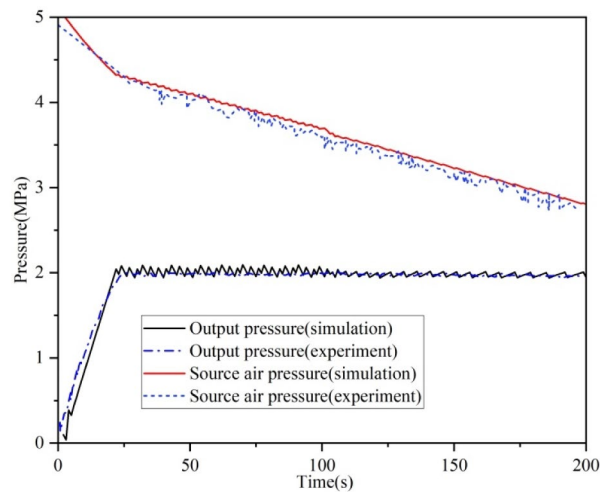
Fig. 5 Experimental equipment of SER [34]

Table 1 Features of prior work vs. our work

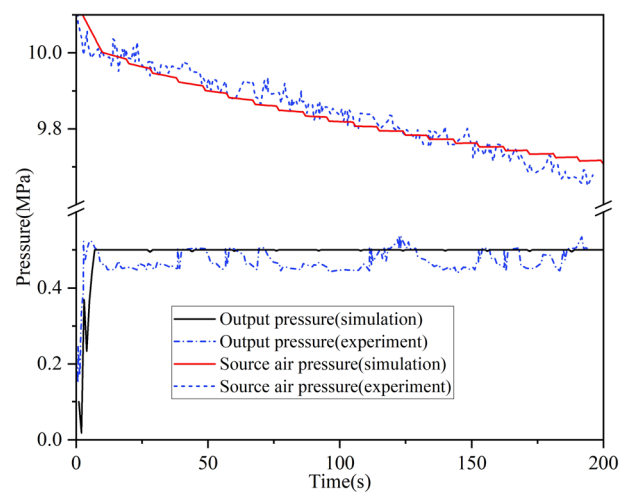
Reference	System Scope	Validation Method	Key Metrics	Optimization/Comparison
Zhao et al. [20]	CAES with throttle valve	Steady-state efficiency analysis	Round-trip efficiency	Sliding pressure/constant pressure
Guo et al. [21]	CAES with ejector	Thermodynamic modeling	Energy storage efficiency	Ejector parameters
Guan et al. [22]	CAES with nozzle	Orthogonal experimental design	Efficiency	Optimal nozzle governing method
Chen et al. [25]	CAES with bypass systems	Thermodynamic modeling	Energy density	Optimizing performance under variable pressure conditions
Guo et al. [26]	CAES with inlet vanes adjustment	Thermodynamic modeling	Exergy efficiency	Inlet vanes adjustment optimization
Our work	CAES with SER	Thermodynamic modeling	Exergy destruction, AGC indicator	SER optimization

Table 2 Test parameters of SER verification model

Parameter	Value	Parameter	Situation 1	Situation 2
Environment pressure	101.3kPa	Initial supply pressure	4.96MPa	10.12MPa
Area of switch valve	7.095mm ²	Target outlet pressure	2MPa	0.5MPa
Supply tank volume	3×90L	Area of outlet choke	7.095mm ²	3.14 mm ²
Expansion tank volume	50L	Output mass flow rate	31g/s	3.5g/s
Initial expansion tank pressure	101.3kPa	Environment temperature	285K	285K



(a) Situation 1



(b) Situation 2

Fig. 6 Simulation result contracted with experiment result

Table 3 Error calculation

Error	Calculation Formula	Parameter	Situation 1	Situation 2
RMSE (MPa)	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (p_{sim,i} - p_{exp,i})^2}$	Source air pressure	0.15MPa	0.056MPa
		Output pressure	0.06MPa	0.029MPa
MAPE (%)	$MAPE = \frac{1}{n} \sum_{i=1}^n \left \frac{p_{sim,i} - p_{exp,i}}{p_{exp,i}} \right \times 100\%$	Source air pressure	2.5%	0.3%
		Output pressure	3.4%	8.7%

Table 4 Design parameters of the TS-CAES discharging system

Parameter	Value	Parameter	Value
Environment temperature	298 K	Isentropic efficiency	0.88
Environment pressure	0.1 MPa	Designed mass flowrate	25.35 kg/s
Storage volume	15000m ³	Expansion inlet pressure	7 MPa
Discharging time	6 h	Overall expansion power	10 MW
Maximum Storage pressure	10 MPa	Expander inlet temperature	336.88K
Storage tank diameter	4m	Expansion tank diameter	4m
Tank wall thickness	0.016m	Tank material	Q345R
Switch/regulation valve area	0.064m ³	Switch/regulation valve C _q	0.95

condition are 10 MW and 6 h with a 4-stage isometric expansion, respectively, thus the energy requirement is chosen as 60 MWh. The simulation sampling time is 1s.

4.1.2 Operating condition design

The overall rated expansion power is 10MW and the expansion inlet pressure remains 7MPa under rate condition.

In order to study the destruction and control characteristics of the system under the regulatory market demand for AGC regulation, the sinusoidal command curve is set. The amplitude and frequency can represent the regulation speed and range of AGC command. The amplitude of the sine command curve is selected as 80% of the rated power [44] of the system(P_0), which is 8MW.

In the regulatory ancillary service market rules, the regulation rate(v_i , MW/min) is taken as an important index of energy storage system AGC assessment [45]. Regulation rate can be calculated by(26):

$$v_i = \frac{P_{before} - P_{after}}{\Delta t} \quad (26)$$

Regulation rate index K_1 is determined and demanded by(27):

$$0.5 < K_1 = \frac{v_i}{v_e} < 5 \quad (27)$$

where, v_e is AGC standard regulation rate determined by industry standard. AGC standard regulation rate of CAES system is 50% P_0 per minute [45]. Combined with the above formula, the system regulation rate is chosen as 2.5MW/min so that the system can meet the regulation demand. The load demand curve is shown in Fig. 7.

4.2 Rated operating

Under rated operating condition, the output mass flow rate and expansion inlet pressure are basically stabilized in 25.35kg/s and 7Mpa. The air pressure and temperature in the expansion tank in SER system are shown in Fig. 8. The pressure control unit can smoothly adjust the outlet pressure of the expansion tank and stabilize it at 7MPa, and the overall change trend of the air temperature in the expansion tank decreases firstly and then increases.

At the beginning of the discharging process, a significant pressure drop at the inlet led to a gradual decrease in the inlet temperature. This resulted in a noticeable decline in the expansion tank's internal temperature. Subsequently, as the inlet temperature gradually recovered and heat transfer between the expansion tank and its surroundings occurred, the air temperature within the tank began to rise slowly.

Exergy change in different components in SER can be seen in Fig. 9. The switch valve and the regulation valve exergy destruction increase with time and the isenthalpic throttling of the switch valve is the main cause for

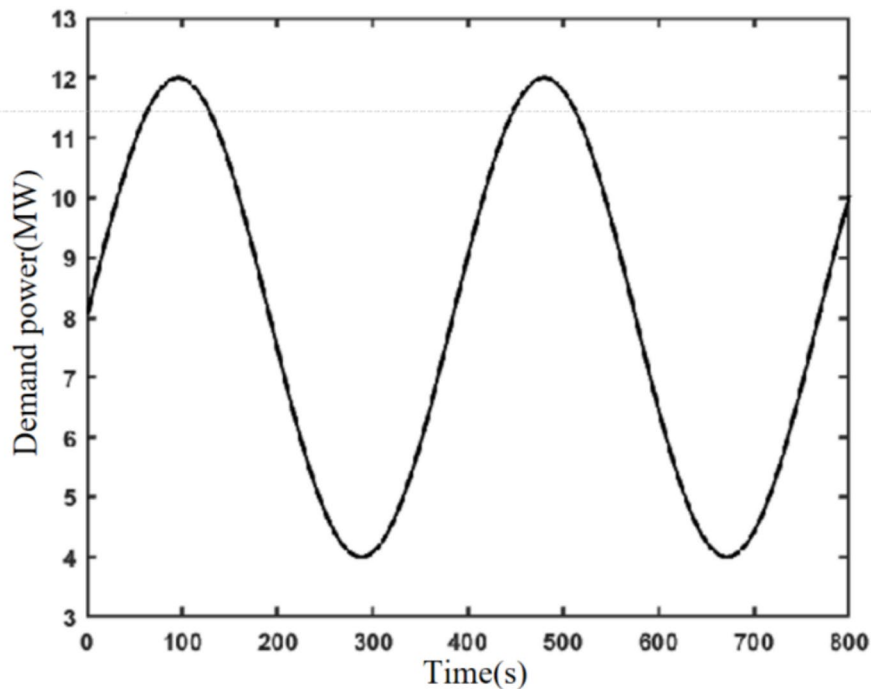


Fig. 7 Load demand curve based on AGC service standard

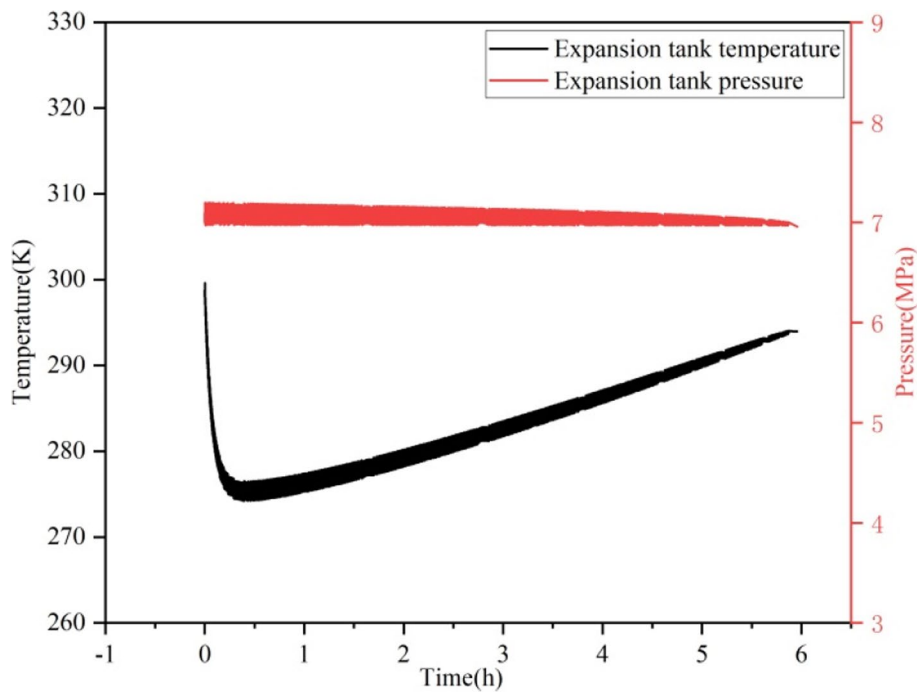


Fig. 8 Variation of the air temperature and pressure in expansion tank

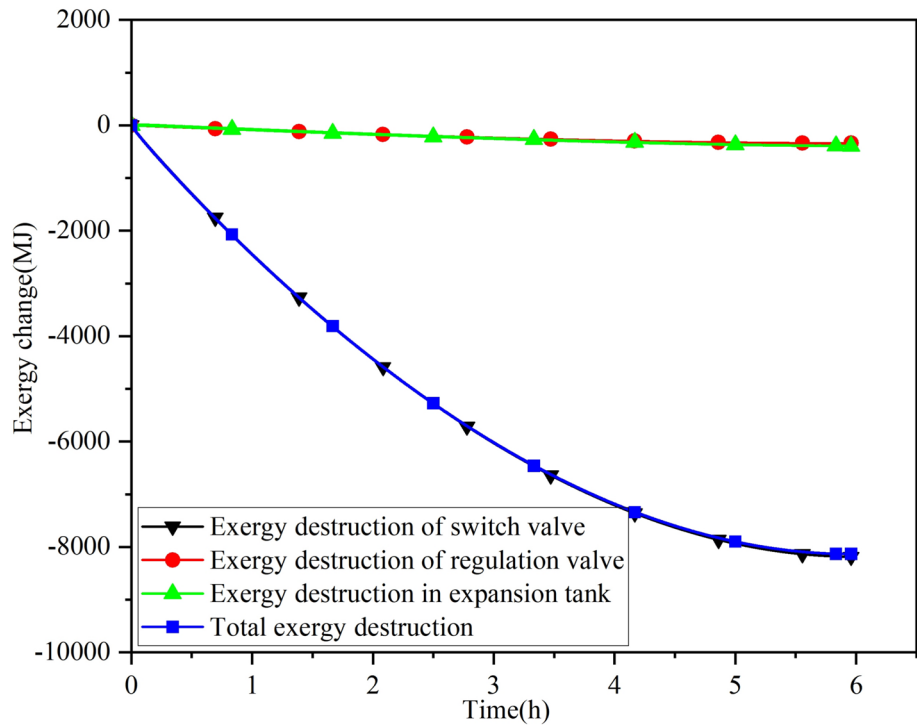


Fig. 9 Exergy change of different SER components in rated operating condition

exergy destruction, for the switch valve having taken most of the pressure drop. Slight exergy increase is in the expansion tank at first, for the obvious temperature gap and heat absorption from the environment. Then the exergy is essentially constant over time. In contrast, exergy destruction of SER unit is slightly lower than that of throttling reduction unit during rated discharging process of the TS-CAES system. Compared with throttling reduction, the exergy destruction is reduced by 3.3%.

4.3 Dynamic operating

Mass flow rate and outlet pressure varying with demand curve in dynamic operating discharging process can be seen in Figs. 10 and 11. During the first 5 h, the mass flow rate and pressure are satisfied with the command curve. Because of the CAES system operating under overload sometimes, the pressure in air storage cavern is not enough to meet the demand in the last 1 h.

The actual pressure and command pressure of expansion tank is shown in Fig. 12. The pressure gap between the tank and command is adjusted by the regulation valve and switch valve. When the expansion tank actual pressure is higher than the command pressure, the regulation valve works to control the outlet pressure and mass flow rate to meet the command; Otherwise, the switch valve works to charge the expansion tank. The regulating role

of the regulation valve is dominate when command pressure drops.

Variation of expansion tank temperature reference to pressure change is shown in Fig. 13. The temperature fluctuates periodically with time due to the variation of the pressure in expansion tank, which is caused by the inlet and outlet flow. Compared with the flow-induced temperature change, the impact of heat exchange with environment is small.

4.3.1 Regulation characteristic

In SER control unit, the combination of a switch valve and a regulation valve is in charge of regulating the inlet mass flow rate and pressure of the expander, using an expansion tank to buffer high pressure from the air storage cavern. The switch valve is installed between the cavern and the expansion tank, bearing the major pressure drop without the flow control requirement, which is simple to operate and requires low accuracy and sensitivity.

The regulation valve is used for the mass flow rate control and further regulation of the pressure by changing the opening area, which has the same regulation principle as throttle valve as mentioned in Fig. 3. In throttle valve-only control unit, the throttle valve should regulate the mass flow rate precisely when bares the pressure drop from air storage cavern to the inlet of expander at the

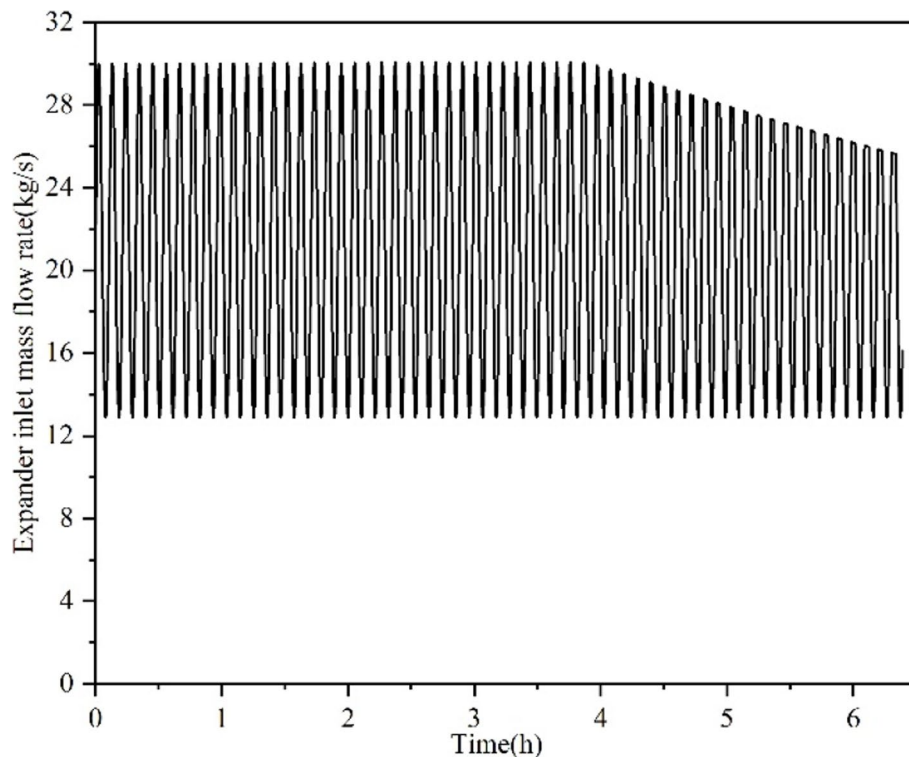


Fig. 10 Variation of output Mass flow rate

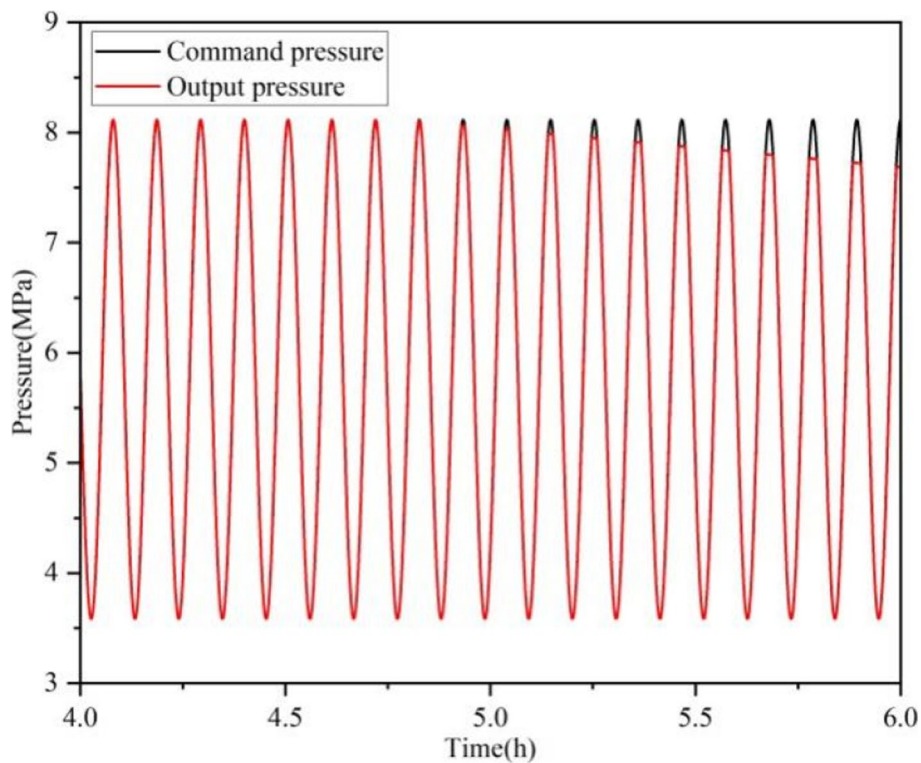


Fig. 11 Variation of the output pressure compared with command pressure

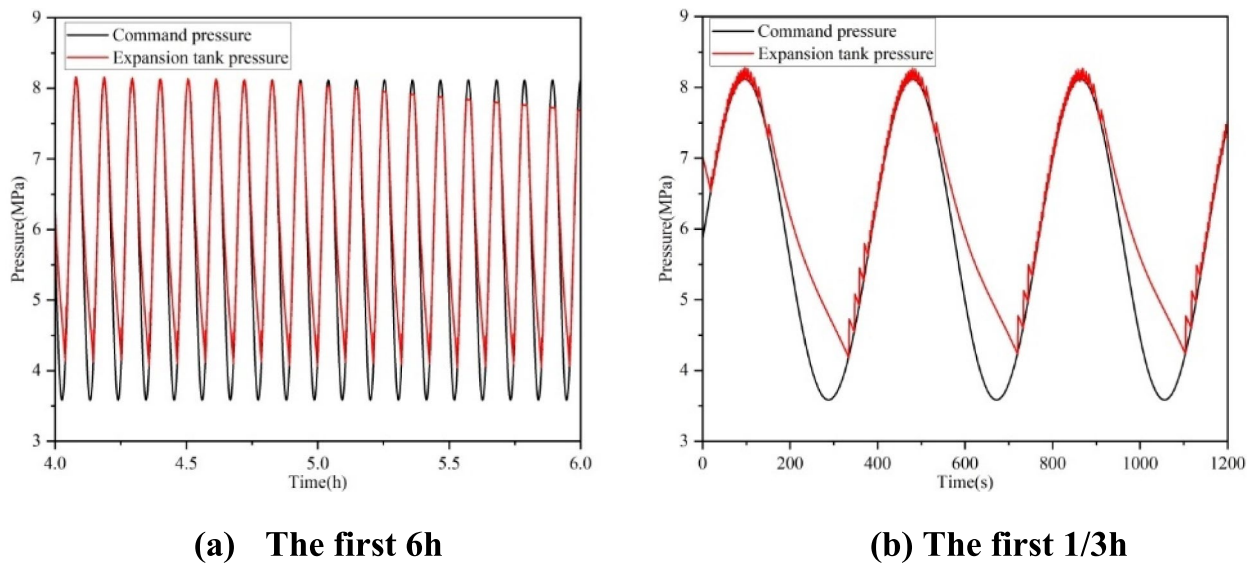


Fig. 12 The pressure and command pressure of expansion tank in (a) the first 6 h; (b) the first 1/3h

same time. In contrast, the pressure difference between the front and back of the regulation valve is smaller because of the decompression function of the switch valve and expansion tank, which can improve the accuracy of mass flow regulation. The control logic of SER and

single-throttle valve and is shown in Table 5 the control logic of SER and single-throttle valve

Under the flow regulation requirements shown in Fig. 14a. To satisfy the expander inlet flow requirement, the mass flow rate varies from 12.9kg/s to 30kg/s

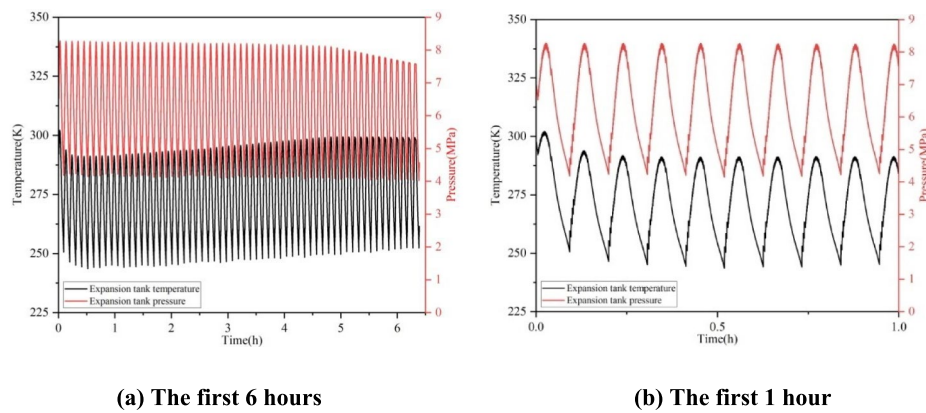


Fig. 13 Variation of expansion tank temperature and pressure in (a) the first 6 h and (b) the first 1 h

Table 5 The control logic of SER and single-throttle valve

Method	Control mechanism	Regulation mode	Control target
SER	Switch valve	On–off	Taking major pressure drop
	Regulation valve	Open area	Regulating mass flow rate
Single valve	Throttle valve	Open area	Regulating pressure and mass flow rate

and the pressure varies from 3.6MPa to 8.1MPa, as the pressure in air storage cavern going down from 10MPa. For the first 5 h, the mass flow rate and pressure successfully match the command curve. Nonetheless, the CAES system's operation under overload conditions results in insufficient air storage cavern pressure to meet the final hour's requirements. The regulation characteristic status of the throttle valve in single-valve regulation and the regulation valve in SER regulation are shown in Fig. 14b–e. The downstream pressure changes from 4MPa–8MPa while the opening ranges over 0.1–0.3. This operation within a high-slope region of the flow-opening curve leads to a high flow rate change rate, predominantly fluctuating within 80–160 kg/s, about 0.8–1.6kg/(s.%). Such high sensitivity means that a minor 1% adjustment in valve opening can trigger significant mass flow variations of 0.8–1.6kg/s, causing difficulties in achieving regulation accuracy and often resulting in overshoot or undershoot during pressure regulation.

In contrast, the regulation valve in the SER system operates within a more reasonable dimensionless opening range of 0.2–0.75, where the flow-opening curve maintains a moderate slope. Correspondingly, its flow rate change rate stabilizes at 30–50 kg/s, about 0.3–0.5 kg/(s.%). This means a 1% opening adjustment triggers a predictable 0.3–0.5 kg/s change in mass flow rate, far less than the 0.8–1.6kg/s change caused by a 1% adjustment

in the single-throttle valve (operating in the 0.1–0.3 opening range). The lower, more stable flow rate change rate enables the SER system to control mass flow and pressure with a less error.

In summary, the SER control unit shows superior regulation strategy which combines simple control method of switch valve with a good operating performance of regulation valve, solving the problem of inaccurate regulation of conventional single-valve throttling.

4.3.2 Exergy change

Taking 100m³ of the expansion tank volume as an example, the exergy change for different components in SER can be seen in Fig. 15. As shown in Fig. 15b, all of the curves fluctuate periodically with the off-design command curve. When the command pressure rises, the mass and pressure in expansion tank increase and the switch valve runs in most of time to charge air into the expansion tank, which leads to exergy destruction; Pressure before and after the regulation valve is basically equal, so that the exergy loss of regulation valve is small. When the command pressure declines, the expansion tank pressure decreases and the regulation valve operates majorly, leading to exergy destruction.

The switch valve contributes most to the exergy destruction in SER system with an exergy loss rate of 83.2%, because the switch valve undertakes the major pressure drop of the air. For ideal gases, exergy is directly

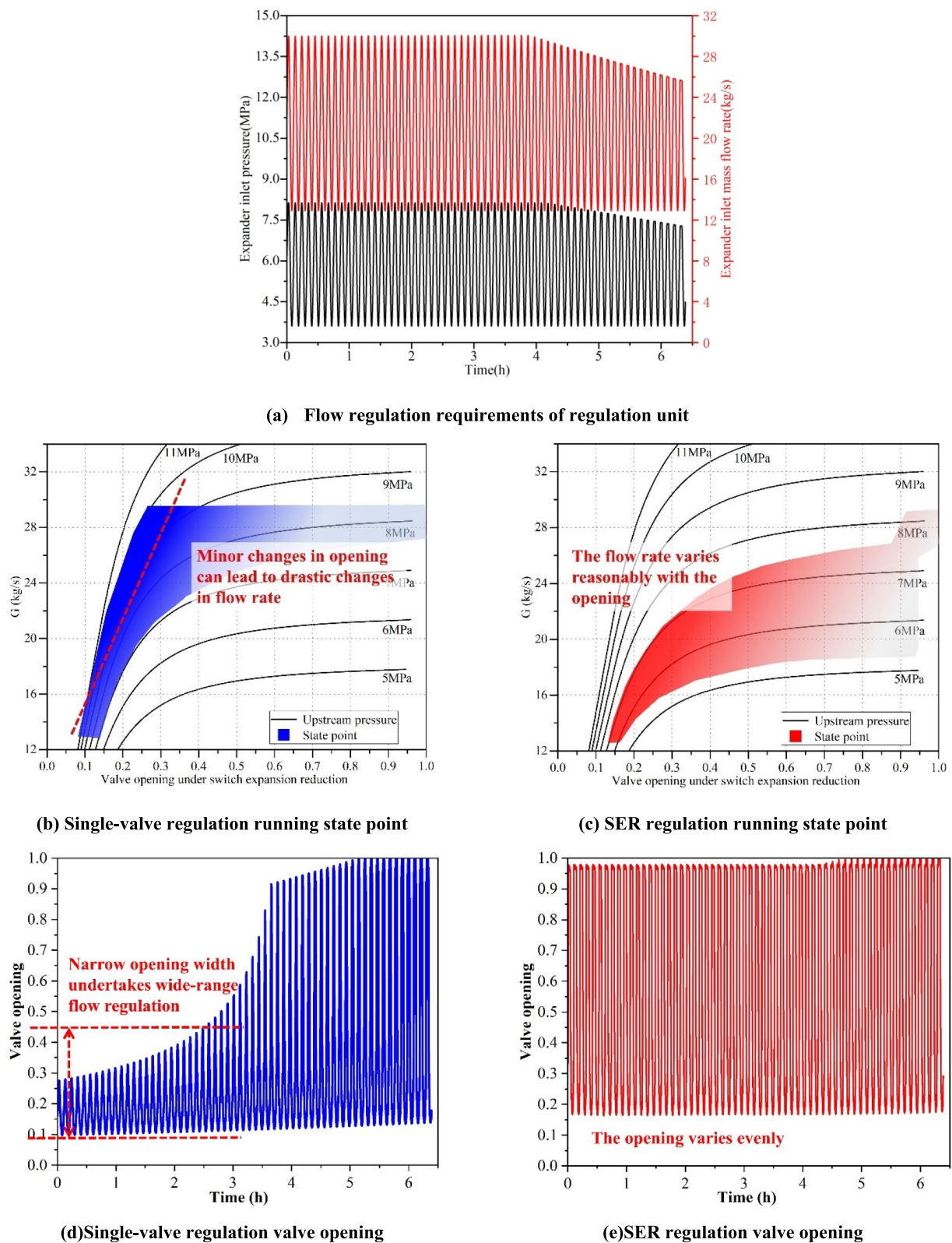
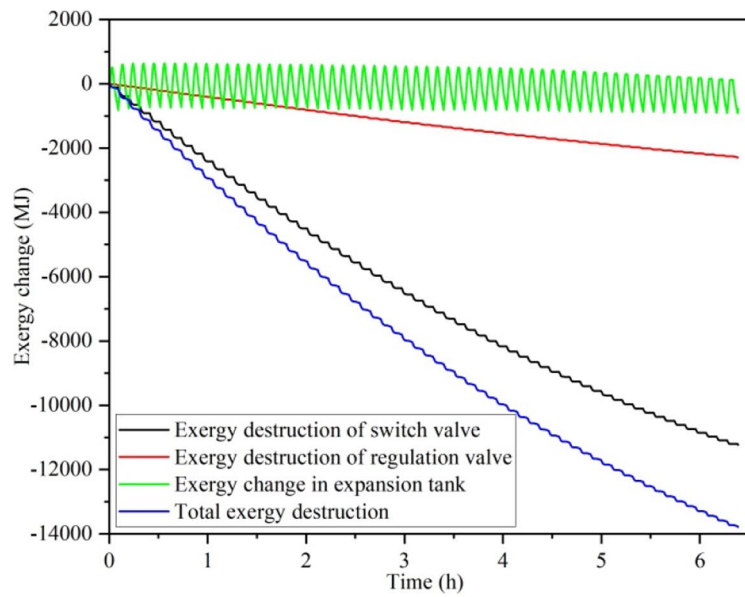
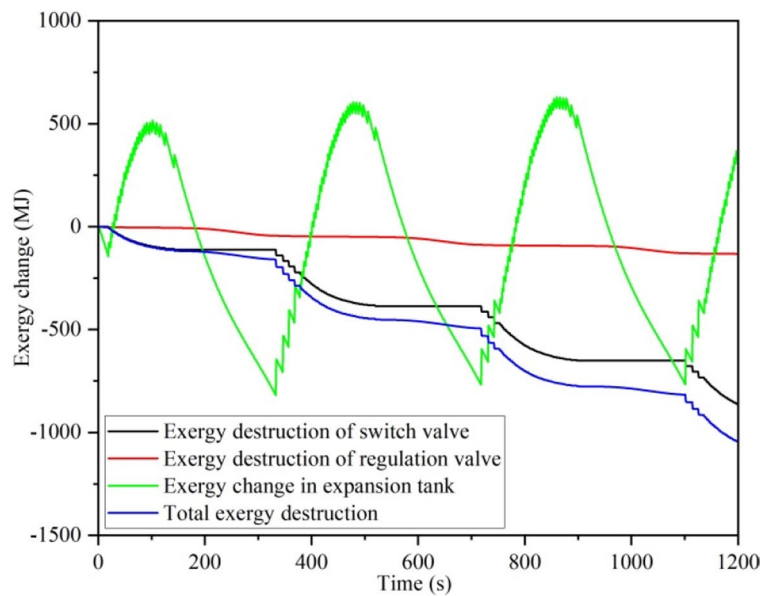


Fig. 14 Pressure and valve opening characteristics of single valve/SER regulation under dynamic operation conditions



(a) the first 6 hours



(b) the first 1200 seconds

Fig. 15 Exergy change variation of different SER components in (a) the first 6 h; (b) the first 1200 s of discharging process

proportional to pressure. The exergy change in the expansion tank fluctuates with pressure and mass in expansion tank, and the heat exchange with the ambient is small. As for the regulation valve, the exergy destruction rate is 16.8%.

The comparison of the exergy destruction of the two regulation methods is shown in Table 6. The SER control

unit is basically more energy efficient than single-valve regulation. In dynamic operating condition, the exergy destruction of SER can be 6.95% of the total internal exergy, which saves 0.52% of the exergy compared to the single-valve case. The additional electrical output per discharge cycle is 270 kWh in the 60MWh system.

Table 6 Comparison of the exergy destruction of the two regulation methods

Condition	Regulation methods	Exergy destruction	Ratio to total input exergy
Rated discharging operation	SER	8133MJ	4.07%
	Single-valve	8407 MJ	4.21%
Dynamic discharging operation	SER	13873MJ	6.95%
	Single-valve	14922MJ	7.47%

Changing the volume of the expansion tank in SER unit, the laws of pressure fluctuation are shown in Fig. 16. The pressure fluctuation in expansion tank becomes smoother when the volume increases, as the in and out mass flow rate determined by pressure differences and commands are almost the same in different volume situation.

Exergy change of different components of SER varying with the expansion tank volume are shown in Fig. 17a-c. The exergy destruction of the switch valve decreases with the rising volume, as shown in Fig. 17a. As the volume is larger and the output mass flow rate is certain, it takes more time to reach the command pressure in decreasing period, which leads to more time the regulation valve works and less time the switch valve works.

For the regulation valve in Fig. 17b, the exergy destruction decreases when the volume increase from 20m³ to 70m³ because the pressure inside the expansion tank changes more sensitively with volume decreasing before 70m³ and the regulation effect is enhanced to reach the expander's demand inlet pressure. When the volume is larger than 70m³, the pressure difference before and after the regulation valve increases and the regulation valve working time is extended, which leads to the rising exergy destruction.

Figure 17c shows the exergy change in expansion tank, because of the air mass increase, the exergy wave gets more intense with volume increasing.

The exergy destruction of SER is calculated by Eq. (25). Figure 18 shows the variation law of total exergy loss of SER with expansion tank volume when discharging process finishes. The ratio of the exergy destruction of the SER to total internal exergy varies from 6.88% to 7.12%. Due to the combined effect of the two valves in different volumes, the exergy destruction is minimized at a volume of 70m³. It can be assumed that there is always an optimal expansion tank volume that makes the least exergy loss of SER unit. Under this given system parameters, 70m³ can be the optimal choice to reduce the irreversible loss in flow-control unit in CAES' dynamic operating condition.

5 Optimal strategy

5.1 Expansion tank volume optimization design method

In order to study the relationship between optimal volume and system parameters under specific dynamic characteristics, two assumptions need to be made about the load sine curve:

- (1) The swing of the command power wave is 80% of the rated power (80%P₀).
- (2) The power climbing rate fits the AGC service requirements, which means the command varies 25% of the rated power per minute (25%P₀).

The design mass flow rate of this four-stage expanders is calculated by:

$$Ge_0 = P_0 / (c_p \cdot \eta_e (1 - \pi^{\frac{1-k}{k}}) \cdot \sum_{i=1}^N T_{ei, in}) \quad (28)$$

where, η_e is the isentropic efficiency; $T_{ei, in}$ represent the inlet air temperature of each stage of expander.

For specific system, the design parameters are certain, the design mass flow rate is calculated by:

$$Ge_0 = \frac{P_0}{M \left[1 - \left(\frac{p_e}{p_0} \right)^{\frac{1-k}{k}} \right]} \quad (29)$$

where, M represents system specific parameter only related to efficiency and temperature designed before.

$$M = c_p \cdot \eta_e \cdot \sum_{i=1}^N T_{ei, in} \quad (30)$$

On the basis of equation of state of ideal gas, the design mass flow rate can be also expressed as:

$$Ge_0 = \frac{V_{cav}(p_e - p_0)}{R_g T_0 t} \quad (31)$$

The cavern volume is in direct proportion to discharging time when the pressure and power condition is certain:

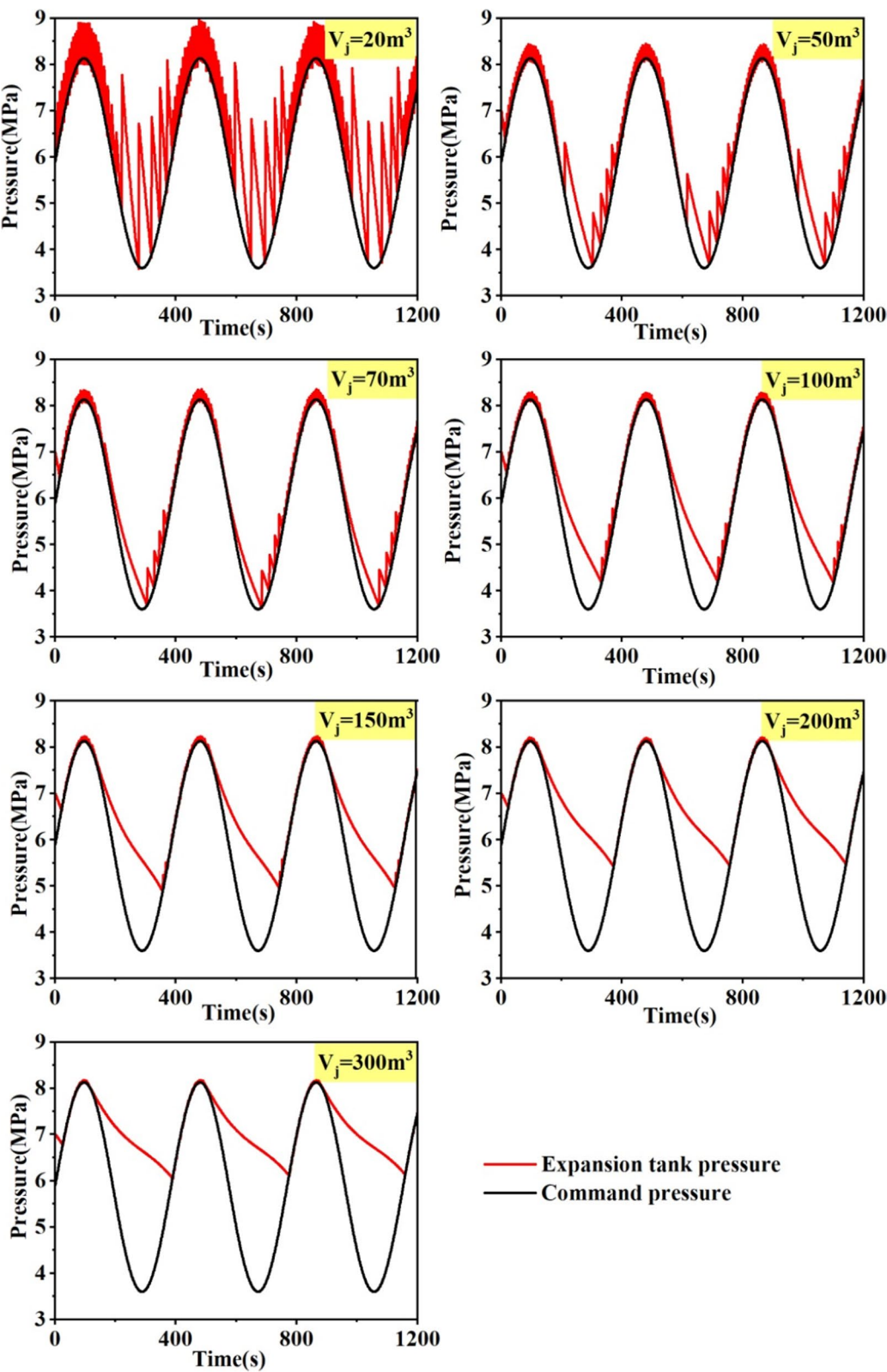
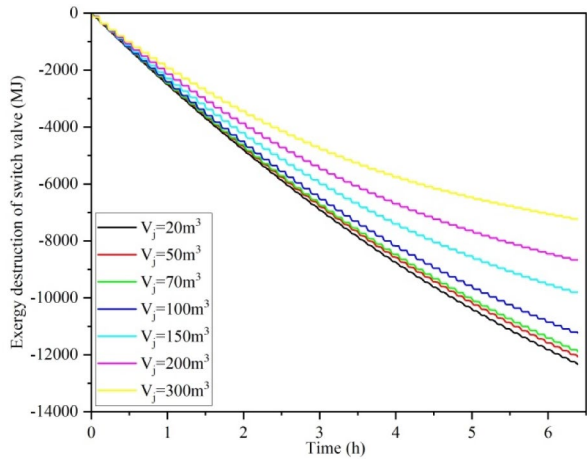
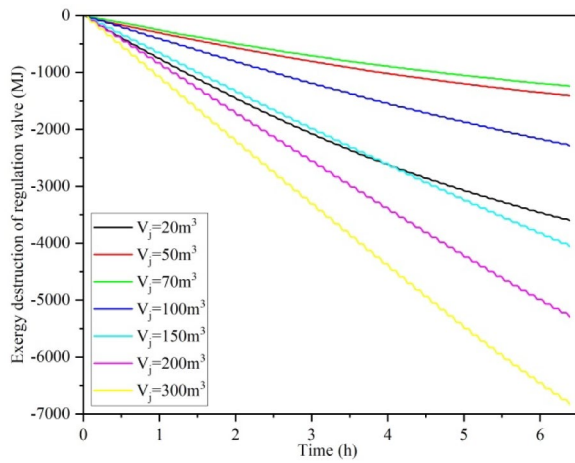


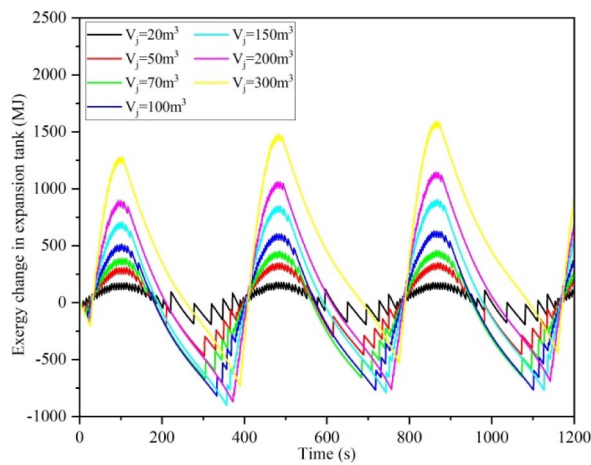
Fig. 16 Variation law of pressure of expansion tank with expansion tank volume



(a) Switch value



(b) Throttle value



(c) Expansion tank

Fig. 17 Variation law of exergy changes of different SER components with expansion tank volume during discharging process

$$V_{cav} = Nt \quad (32)$$

Based on Eqs. (28)-(32), the optimal expansion tank volume estimation value can be calculated by:

$$V_{j,optimal} = \frac{\beta P_0 R_g T_0}{M(p_e - p_0) \left[1 - \left(\frac{p_e}{p_0} \right)^{\frac{1-k}{k}} \right]} \quad (33)$$

where, β is a scaling coefficient that balances the expansion tank's pressure buffering capacity and the system's dynamic flow demand. Its physical meaning lies in quantifying the matching ratio between the expansion tank's volume (for suppressing transient flow fluctuations) and the system's base flow intensity (V_{cav}/t , reflecting the average discharge rate of the air storage cavern).

$$\beta = \frac{V_{j,optimal}}{V_{cav}/t} \quad (34)$$

According to this simulation results, the cavern volume has little influence to the expansion tank volume. When the volume(V) is 15000m³ and discharging time(t) is 6 h, the optimal volume of expansion tank ($V_{j,optimal}$) is 70m³, which leads to:

$$\beta = \frac{V_{j,optimal}}{V_{cav}/t} = \frac{1}{25} \quad (35)$$

For the 10 MW/60 MWh system, $\beta=1/25$, indicating the optimal expansion tank volume only needs to account for 4% of the base flow intensity—this ratio arises because the expansion tank serves short-term buffering rather than long-term energy storage.

As for other application situations with different load varying frequency, it can be considered that the load frequency does not affect the relationship between the system parameters and the expansion tank. The optimized volume prediction formula can be expressed as:

$$V_{j,optimal} = \frac{\beta P_0 R_g T_0}{M(p_e - p_0) \left[1 - \left(\frac{p_e}{p_0} \right)^{\frac{1-k}{k}} \right]} \cdot f(F) \quad (36)$$

$f(F)$ is used to represent the effect of frequency on the optimized volume.

Based on the formula, an optimal volume of the expansion tank can be roughly determined, and the optimization direction can be judged. The optimal expansion tank volume increases with the increase of power level; The initial pressure of the cavern increases and the volume of the expansion tank decreases.

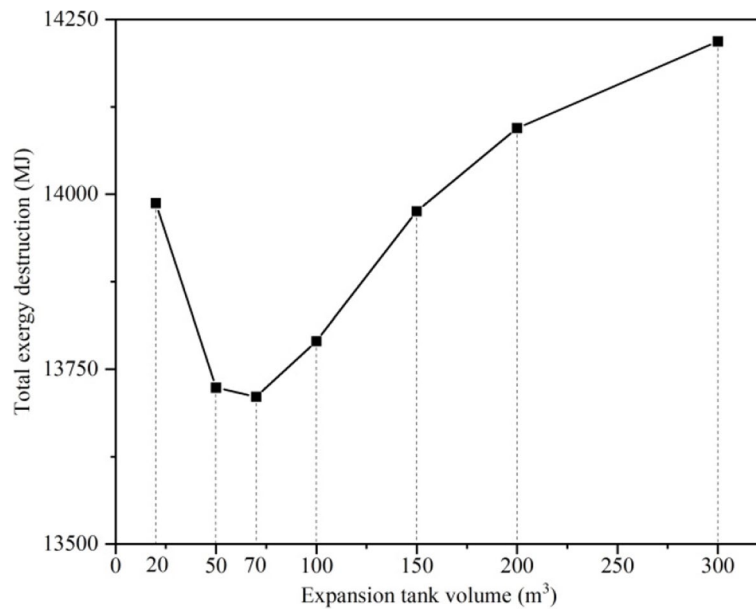


Fig. 18 Variation law of total exergy loss of SER with expansion tank volume

5.2 Optimization of the expansion tank pressure control strategy

The opening and closing of the switch valve are controlled by the expansion tank pressure signal obtained from sensor. In present strategy proposed by Li et al. [36], when the pressure in the expansion tank is greater than the demand inlet pressure, the switch valve is closed, and the opening of the regulation valve is adjusted so that the pressure behind the valve is equal to the command pressure. On the contrary, the switch valve is opened, the air in the air storage cavern flows into the expansion tank, and the opening of the regulation valve is fully opened, so as to meet the command as much as possible. Considering that the switch valve action takes time, if the command pressure changes rapidly or the switch valve flow area is large, the above control strategy could lead to mismatch with the command or abnormal switching.

In the CAES system, both the number of switch valve operations and the stability of pressure curve variations are critical factors affecting system stability. Decreasing the number of switch valve actions and ensuring stable pressure curve changes contribute significantly to enhancing system stability.

To optimize the control strategy, adjustment width can be proposed instead of single curve. Figure 19 gives the proposed control flow chart of SER system. The adjustment width refers to the pressure interval between the upper command pressure and lower command pressure, serving as a pressure buffer zone to stabilize valves operations. The detailed control strategy is

listed below. When the pressure in the expansion tank is greater than the upper command pressure p_U , the switch valve is closed and the opening of the regulation valve is adjusted; When the pressure in the expansion tank is lower than the lower command pressure p_L , the switch valve is opened, and the opening of the regulation valve is fully opened.

The lower and upper command pressure are supposed to ensure the switch value running stability and meet the adjustment accuracy.

Firstly, the pressure adjustment width is supposed to wider than the maximum pressure change under the switch valve response time (t_s), which ensures that the switch valve has enough time to complete the action when increasing the sample frequency. The minimum pressure adjust width ($\Delta p_{\min}$) can be calculated by:

$$\Delta p_{\min} = p_U - p_L = t_s \left(\frac{dp_j}{dt} \right)_{\max} \quad (37)$$

Based on the estimation of the pressure rise of the expansion tank and the response time of the switching valve [46] in this system, the pressure change is supposed to be wider than 0.15bar.

When the expansion tank pressure is lower than the demand pressure, the adjustment error is determined as the difference between expansion tank pressure and demand pressure. According to the regulation accuracy requirements of the frequency modulation market, the allowable error of the regulation is 1.5% of the rated output power. In this paper, the adjustment error is set as 1.05bar under given characters, which means:

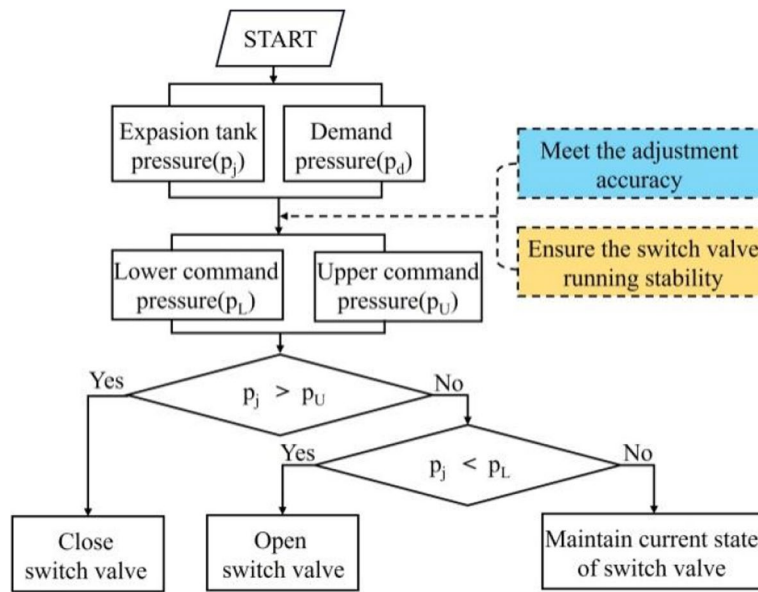


Fig. 19 Optimal calculation logic diagram of TS-CAES system discharging process based on SER

$$p_L = p_d - 1.05 \text{ bar} \quad (38)$$

Figure 20 compares the simulation curves when the adjustment width is set in 2 bar and 3 bar. Figure 21 shows the on and off times of the switch valve. The expansion tank pressure is more stable when the adjustment with is 3 bar rather than 2 bar, which means that the increase in adjustment width make the switch value run smoothly and on–off frequency is reduced. Figure 22 shows the frequency of switch valves and the exergy destruction varied with the adjustment width. Combined with Fig. 21, when the adjustment width is 3 bar, the exergy destruction increase is lower than 1.5% and the frequency is 50% decreased, meanwhile, the on–off action time is more evenness.

Increasing the adjustment width can effectively reduce switching valve operations while stabilizing pressure curve fluctuations. As shown in Figs. 19, 20, and 21, a 3-bar width decreases switching frequency by 50% while stabilizing pressure fluctuations within ± 1.2 bar. However, excessively wide widths (e.g., > 3 bar) create unnecessary pressure differences between the expansion tank and expander inlet, increasing throttling losses across the regulation valve, confirming that 3 bar is the pragmatically optimal choice for dynamic conditions.

6 Conclusion

In this paper, a thermodynamic modelling of a 10 MW CAES system with thermal storage integrated pressure control unit is established. The discharging process of TS-CAES system based on SER is proposed, which is

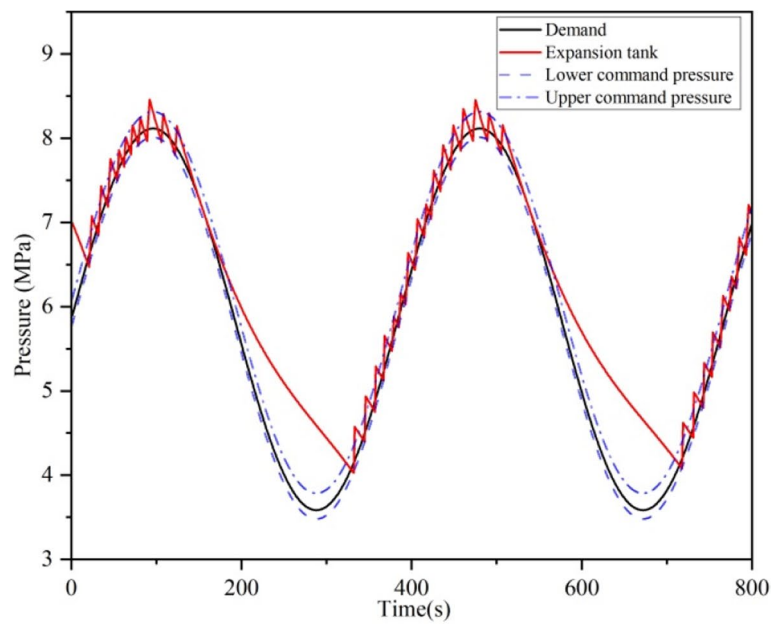
effective for control accuracy and exergy loss comparing with single-valve throttle regulation unit. Modelings are established and validated by experimental results. Taking 10 MW TS-CAES discharging system as an example, the operation law and control method of the simulation unit are investigated under the dynamic application background of participating in secondary frequency modulation service. The main conclusions are as follows.

Under dynamic operating condition, the SER control unit demonstrates outstanding regulation performance in terms of accuracy and efficiency by decoupling the pressure control and mass flow control.

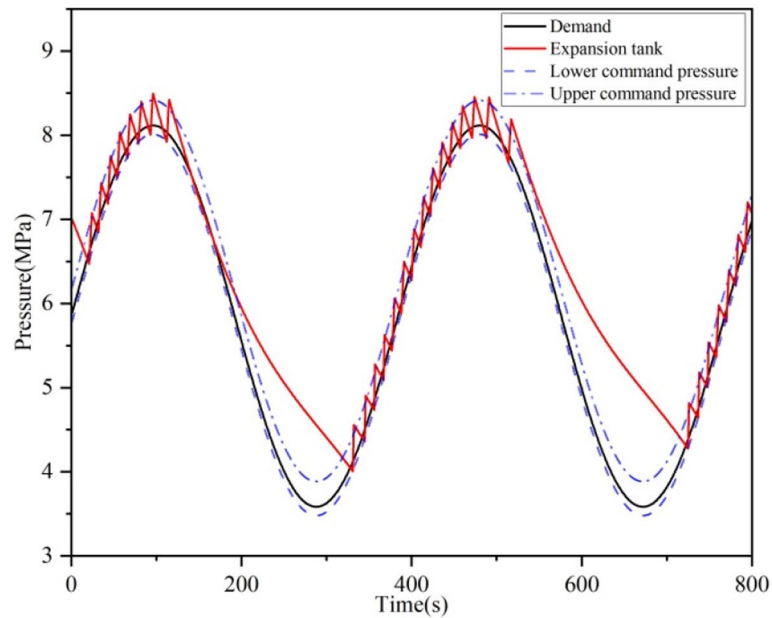
Exergy analysis of the whole SER unit and components varied with expansion tank volume is proposed, indicating that there is always an optimal volume to minimize the exergy destruction because of the combination of two valves.

Based on dynamic simulation results and thermodynamic formulas, a qualitative formula for the relationship between optimal volume of the expansion tank in SER and system parameters is derived, which can help judge the optimization direction of expansion tank volume. Under the simulated 10 MW/60 MWh system parameters and dynamic load to meet the requirements of the unit to participate in the frequency modulation auxiliary service market, the optimal expansion tank volume is about 70 m³.

The optimization method of control strategy of SER system is explored and the control strategy of setting adjustment width instead of single instruction curve



(a) Adjustment width set in 2 bar



(b) Adjustment width set in 3bar

Fig. 20 Variation curve of expansion tank pressure in different adjustment width

is proposed. Furthermore, the method of determining upper and lower limit of instruction interval and adjusting width is established according to the adjustment precision and stability index of switch valve.

Under simulated conditions, when the adjusting width is 0.3Mpa, the operating frequency of the switch valve is stable, and the influence on the adjustment accuracy and energy loss is minimal.

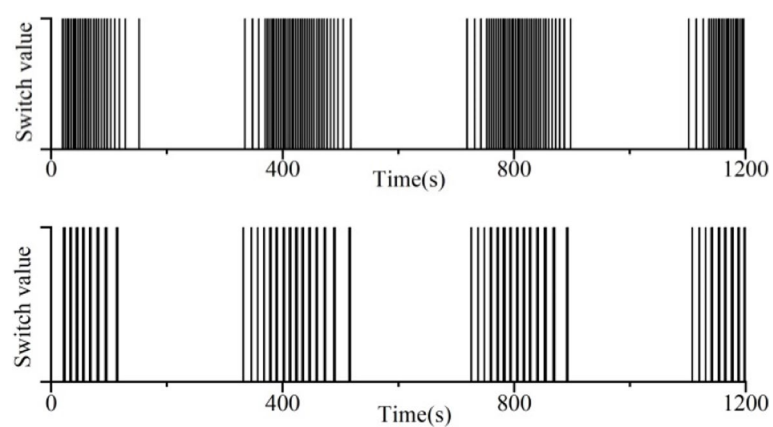


Fig. 21 On–off situation of switch valve under different adjustment width. (upper: 0 bar; lower: 3 bar)

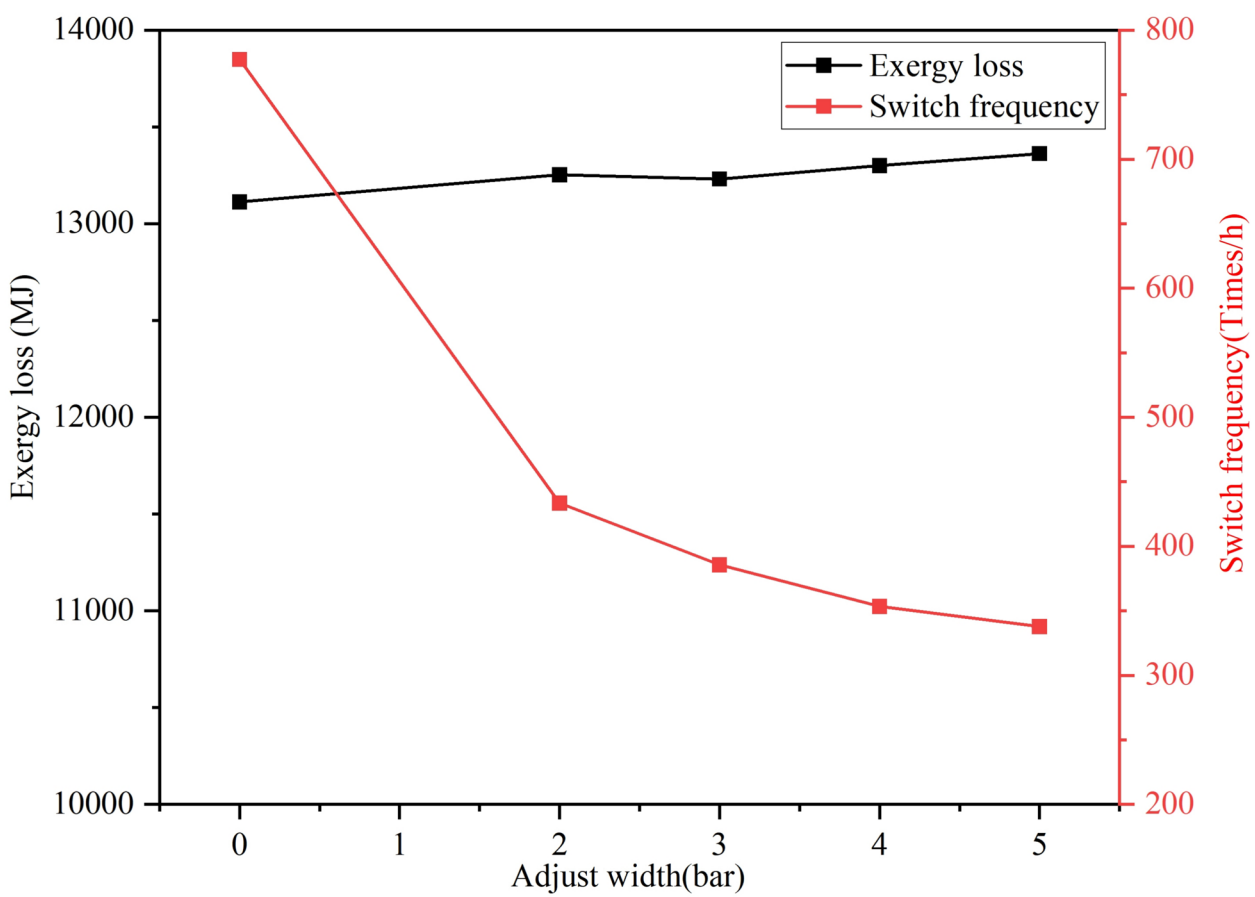


Fig. 22 Variation of exergy loss and switch frequency with adjustment width

7 Nomenclature

Symbols

A: Heat transfer area, m^2
 c: Local sound velocity, m/s
 c_p : Specific heat at constant pressure, $kJ/(kgK)$
 c_v : Specific heat at constant volume, $kJ/(kgK)$
 C_q : Flow coefficient
 Ex: Exergy, kJ
 $f(F)$: Frequency impact factor
 $\dot{G}$: Relative reduced mass flow rate
 G: Air mass flow rate
 h: Specific enthalpy, kJ/kg
 k: Air specific heat capacity ratio
 K_1 : Power regulation rate index
 m: Mass, kg
 $\dot{m}$: Mass flow rate, kg/s
 $\dot{n}$: Relative reduced rotation speed
 n: Rotation speed, rpm
 NTU: Number of heat transfer units
 P: Power, MW
 p: Pressure, MPa
 R_g : Ideal gas constant, $kJ/(kgK)$
 S: Valve effective cross section area, m^2
 t: Time, s
 t_s : Switch valve response time, s
 T: Temperature, K
 U: Heat transfer coefficient, $W/(m^2K)$
 u: Specific internal energy, kJ/kg
 V: Volume, m^3
 v: Specific volume, m^3/kg
 v_i : System power regulation rate, MW/min
 v_e : AGC standard power regulation rate, MW/min
Subscripts
 0: Designed value
 1: Upstream of valve
 2: Downstream of valve
 av: Air vessel
 after: After power change
 before: Before power change
 cav: Air storage cavern
 cold: Cold fluid
 d: Demand pressure
 e: Expander
 env: Environment
 h: Heat exchanger
 hot: Hot fluid
 i: Stage
 in: Inlet flow
 ini: Initial state
 j: Expansion tank
 L: Lower command pressure
 min: Minimum value

max: Maximum value
 optimal: Optimal value
 out: Outlet flow
 reg: Regulation valve
 sw: Switch valve
 tank: Water tank
 U: Upper command pressure
 valve: Valve
Greek symbols
 η : Isentropic efficiency
 $\dot{\eta}$: Relative efficiency
 ε : Effectiveness of heat exchanger
 χ : Ratio of heat capacities of two fluids
 β : Volume coefficient
 ν : The valve operating pressure ratio
 ν_c : Critical pressure ratio
 π : Expansion ratio

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Authors' contributions

SZ: Methodology, Data curation, Investigation, Writing – original draft, Software, Visualization. XZ: Supervision, Conceptualization, Writing – review & editing. RW: Formal analysis. YJ: Methodology, Writing—Review & Editing. YY: Formal analysis. SH: Formal analysis. YX: Project administration. Haisheng Chen: Project administration, Writing—Review & Editing. All authors have read and approved the version published.

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Data availability

All data generated or analysed during this study are included in this published article.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

All authors agree to the publication of this manuscript.

Competing interests

HC is an editorial board member for Carbon Neutrality and was not involved in the editorial review, or the decision to publish this article. All authors declare that there are no competing interests.

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